

ANALYZING THE BUSINESS USING ARCHITECTURES

*Achieving Business Value with
End to End Architecture
Part 1 of 4*

This is a KCI Workshop



Workshop Objectives

- Understand the E2EA architecture approach
- Apply basic architecture analytics to business problems
- Trace architecture impacts from external environment to execution
- Identify hidden relationships
- Understand the value of analytics with architecture

Workshop Structure



Part 1 - The eXternal Architecture (XA)



Part 2 - Strategic Business Architecture (SBA)



Part 3 - Business Architecture (BA)



Part 4 - Enterprise Architecture (EA)

Section 1 – Agenda

What are the 4 architecture perspectives

Some background

The External Architecture (XA)

Analyzing the business landscape

Outcomes and deliverables from the external perspective

Two things to learn in the workshop

**Why should you know about
different architecture perspectives?**

**Why should you pay attention to
analytics?**

Different architecture perspectives:

For example: Operationally you may start with something like basic process and data

BASIC CHOCOLATE CHIP COOKIES

1 cup margarine or butter, softened
3/4 cup granulated sugar
3/4 cup packed brown sugar
1 egg
2 1/4 cups all-purpose flour
1 teaspoon baking soda
1/2 teaspoon salt
1 cup coarsely chopped nuts
1 pkg. (12 ounces) semisweet chocolate chips

Heat oven to 375°. Mix margarine, sugars and egg. Stir in flour, baking soda and salt (dough will be stiff). Stir in nuts and chocolate chips. Drop dough by rounded teaspoonful about 2 inches apart onto un-greased cookie sheet. Bake until light brown, 8 to 10 minutes. Cool slightly then remove from cookie sheet.

Management may want you to use a best practice

Perfect Chocolate Chip Cookies

2 ¼ cups bleached all-purpose Flour
1 tsp. Baking Powder
½ tsp. Baking Soda
2 large Eggs
1 tsp. Vanilla
¼ tsp. Salt
14 Tbsp. Butter (2 sticks minus 2 Tbs.), cut into chunks
¾ cup dark brown Sugar
¾ cup granulated Sugar
2 Tbs. flavorless Oil, such as vegetable or canola

Mix flour, baking powder and baking soda in a medium bowl; set aside. Mix eggs, vanilla and salt in a small bowl; set aside. Microwave butter on high power until just melted but not hot, 30 to 45 seconds; set aside. Mix brown and granulated sugars in a large bowl. Add butter and oil; stir until smooth. Stir in chocolate and optional nuts. Using a 1 ½ ounce (3Tbs) ice cream scoop, spoon 16 dough balls onto a pan that will fit in your freezer. (Don't worry if the dough balls are crowded. They pull apart when frozen.) Freeze until dough is hard, about 30 minutes. (Once dough balls are frozen, they can be stored in freezer bags up to 3 months and baked as desired.)

That's not all, they want options, changes to what the business does

Variations of Chocolate Chip Cookies

- Here are **two choices** each with a different result, one with and one without nuts:
 - 1 ½ cups chocolate chips or 8 ounces good-quality bittersweet or semi-sweet chocolate cut into ¼ inch chunks, about 1 ½ cup
- OR
- 1 cup each chocolate chunks or chips and 1 cup toasted nuts (pecans, walnuts, unsalted peanuts or macadamias)

Then they strategically acquired an older company that makes cookies...

Their terminology may be different (This is where chocolate chip cookies came from)

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"THE SETTLEMENT" COOK BOOK.

CHOCOLATE COOKIES.

½ cup butter,
1 cup sugar,
1 egg,
¼ teaspoon salt,
2 squares (2 oz.) chocolate,

2½ cups flour (scant),
2 teaspoons baking powder,
¼ cup milk.

Cream the butter, add sugar gradually, egg well beaten, salt and chocolate (melted). Beat well and add flour, mixed and sifted with baking powder alternately with milk. Chill, roll very thin, then shape with a small cutter, first dipped in flour, and bake in a hot oven.

Then they tell you to contact process engineering....

Why Engineers Don't Write Cookbooks

Recipe for Chocolate Chip Cookies:

Ingredients:

- 1) 532.35 cm³ gluten
- 2) 4.9 cm³ NaHCO₃
- 3) 4.9 cm³ refined halite
- 4) 236.6 cm³ partially hydrogenated tallow triglyceride
- 5) 177.45 cm³ crystalline C₁₂H₂₂O₁₁
- 6) 177.45 cm³ unrefined C₁₂H₂₂O₁₁
- 7) 4.9 cm³ methyl ether of protocatechuic aldehyde
- 8) Two calcium carbonate-encapsulated avian albumen-coated protein
- 9) 473.2 cm³ theobroma cacao
- 10) 236.6 cm³ de-encapsulated legume meats (sieve size #10)

Steps:

To a 2-L jacketed round reactor vessel (reactor #1) with an overall heat transfer coefficient of about 100 Btu/F-ft²-hr, add ingredients one, two and three with constant agitation. In a second 2-L reactor vessel with a radial flow impeller operating at 100 rpm, add ingredients four, five, six, and seven until the mixture is homogenous.

Questions you should be asking yourself about architecture

These recipes represent all 4 of the architecture perspectives

- What will be the direction from management and what is the impact?
- How similar or different are these?
- Should we adopt the best practice?
- Which version is the one management wants?
- And so on

Well, you may want to apply some analytics but you need to be careful



Number of people killed by dogs

correlates with

Online revenue on Black Friday

Juvenile drug arrests (US)

correlates with

Number of kids killed by their parent

Marriage rate in New York

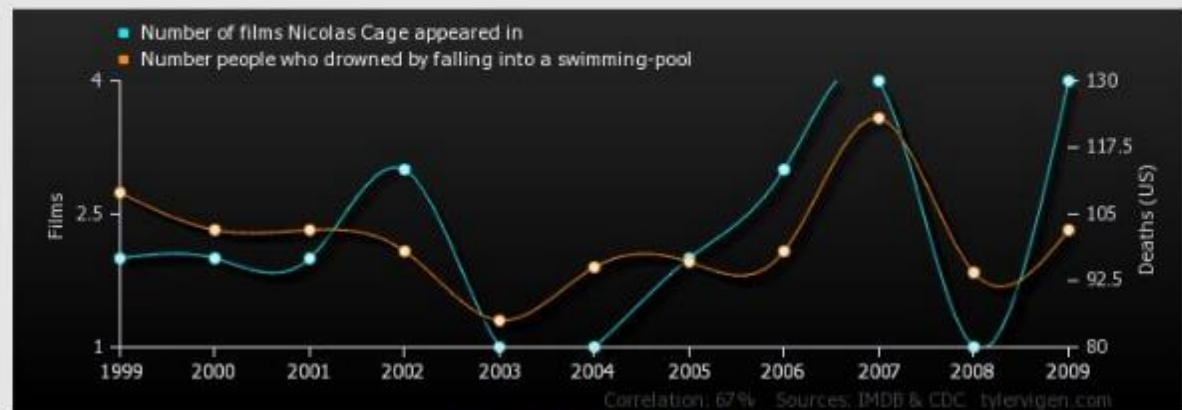
correlates with

Murders by blunt object

Number of films Nicolas Cage appeared in

correlates with

Number people who drowned by falling into a swimming-pool



From the
website Spurious
Correlations

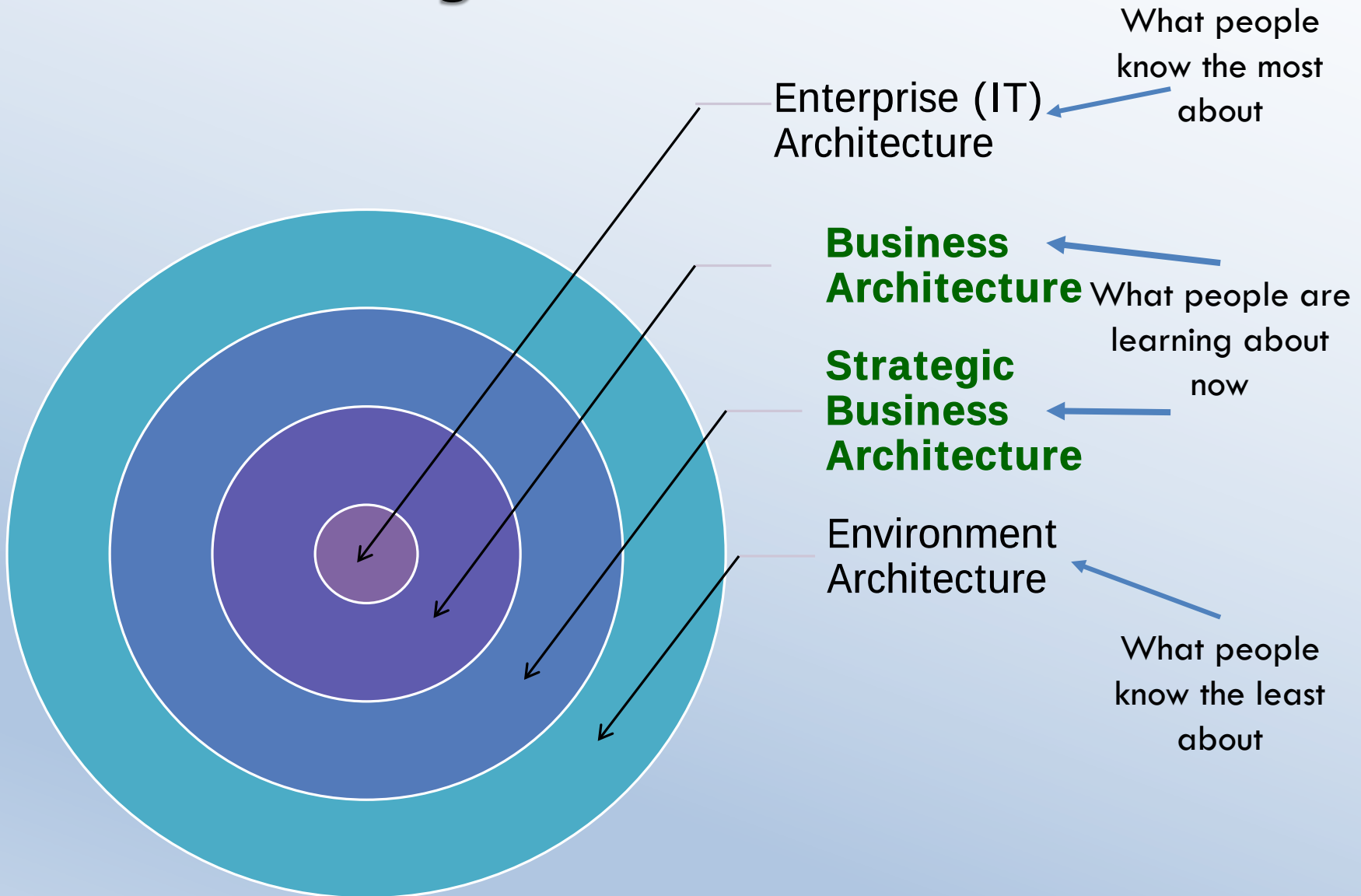
Questions you should be asking yourself about analytics

**None of the correlations imply
cause and effect ...**

**You need to understand
what the data means**

- Is this the data you want to look at?
- What does the algorithm really tell you?
- What if you change the variables?
- And so on... after all it is just data

Architectures have relate to each other through context...



Each architecture perspective is made up of the following parts



Components that are related to in some manner



Models that describe the relationships and provide conclusions



Analytics that work on and in the models that allow for various types of quantitative and qualitative analysis



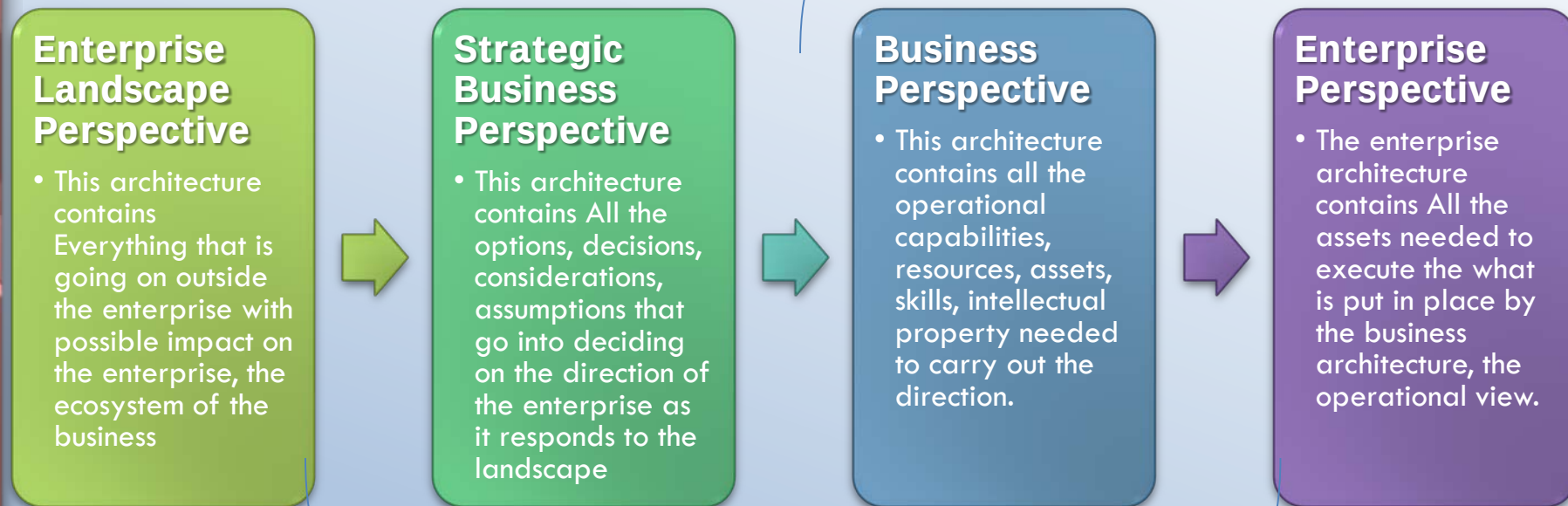
Methodology that delivers the models that have 'answers' related to the results of the models



Body of knowledge that supports interpreting the results of model analysis supporting suggestions for organization/business direction alternatives

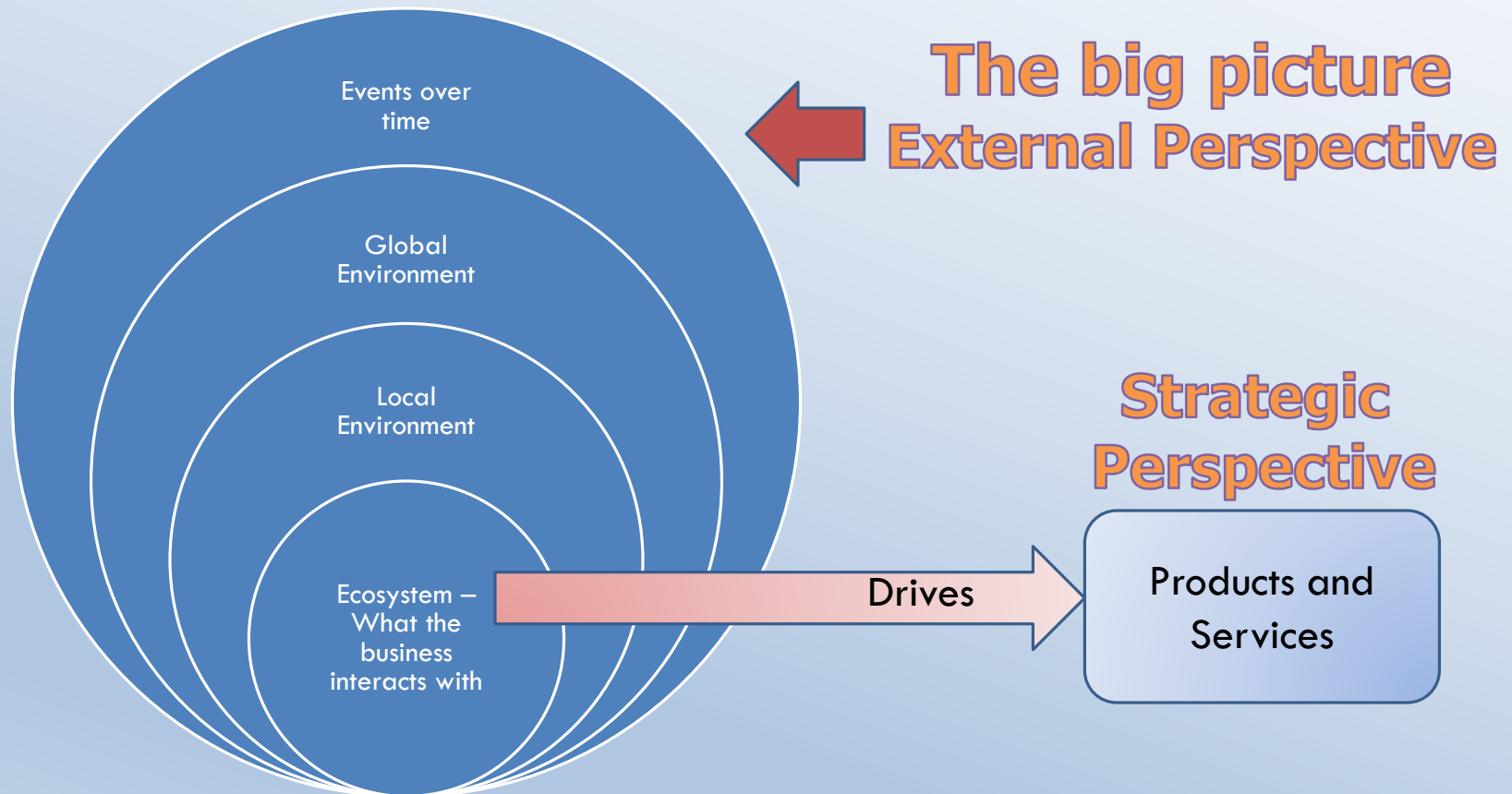
The Relationship of Architectures

Most Organizations do something here

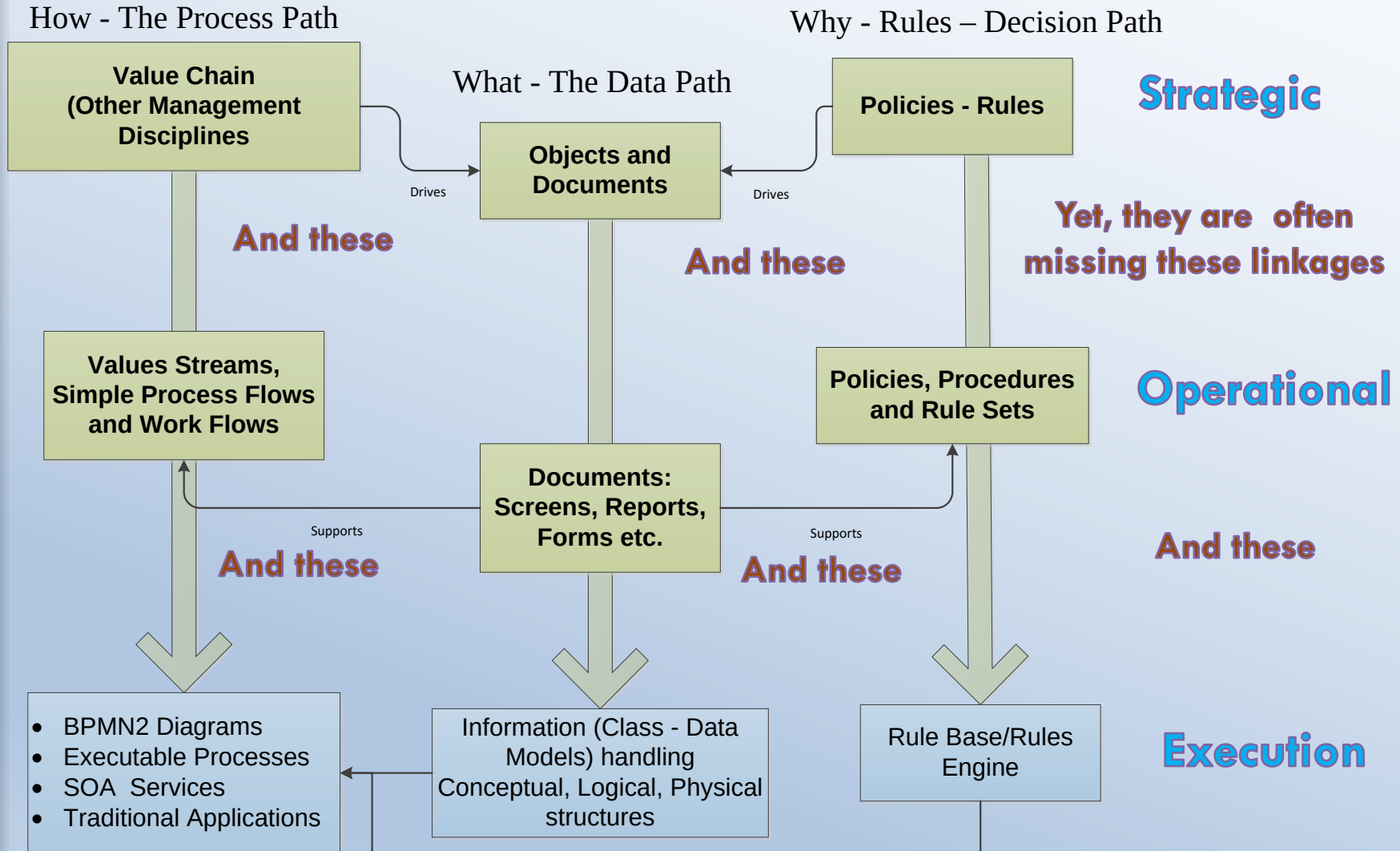


Linkage Analytics Connect These

The external perspective is linked using products and services as related to the business environment



There are 3 major internal paths for linking Strategic, Business and Enterprise Architectures within the business



Let's consider some characteristics of architectures

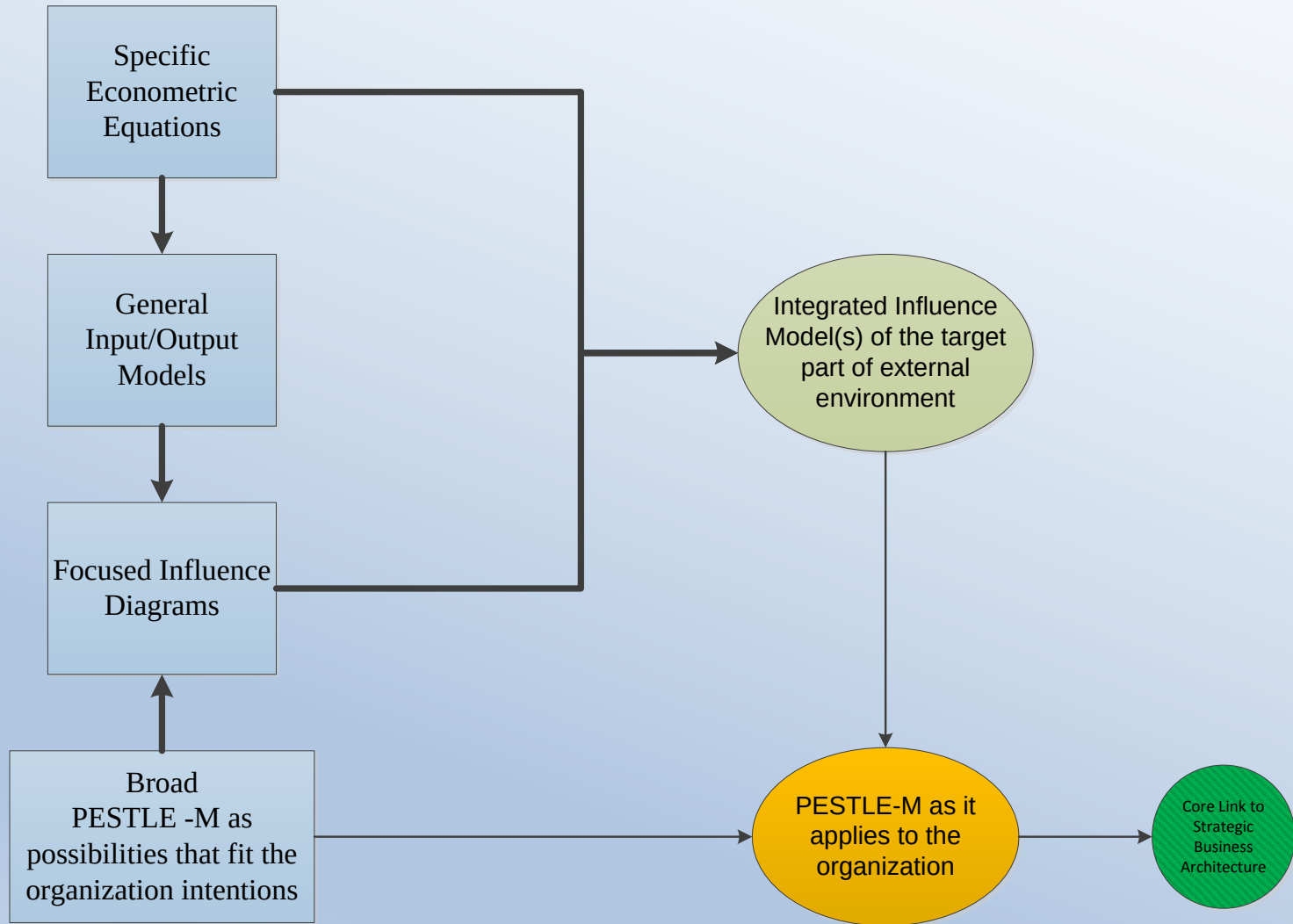
- Components
 - (Lists of things of importance)
- Models
 - (Structural ways to organize things)
- Analytics
 - (Algorithms typically used to examine content)
- Frameworks, Methods and Tools:
 - (Guidelines, step-by-step activities, deliverables, and supporting tools)

Some specifics:

- **Key artifact constructs and components for input to analysis:**
 - E.g. Economic models, business dynamics models, influence diagrams etc.
- **Models for analysis:**
 - E.g. Relationship Matrices, World Dynamics models, econometric models etc.
- **Analytics:**
 - E.g. Qualitative and Quantitative
- **Frameworks, Methods and Tools:**
 - PESTLE, 5 – Forces, SWOT analysis etc.

External landscape architecture has a body of knowledge

The eXternal Architecture - typical components



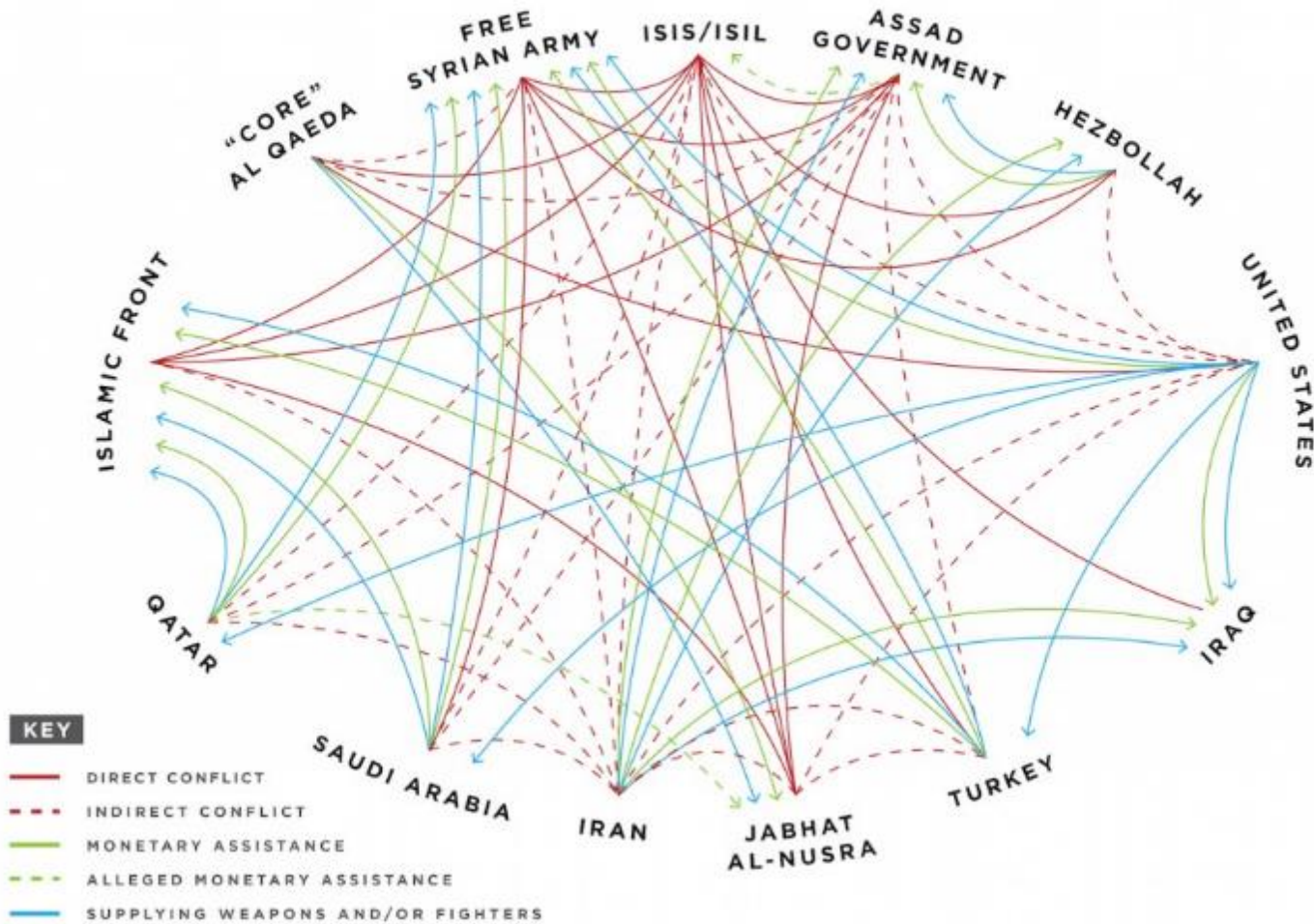
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PA Consulting
Group
PA Knowledge Limited 2009



Another example...

THE TANGLED WEB IN THE FIGHT FOR SYRIA'S FUTURE

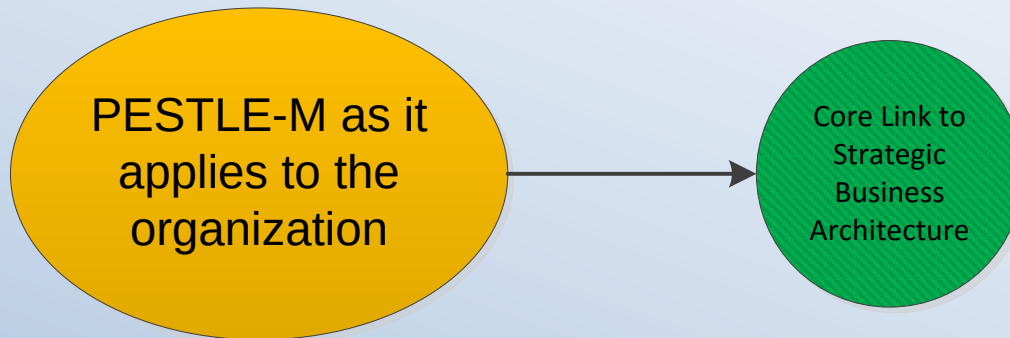


Read the whole story and view sources at [thinkpr.gs/1eVhC14](http://thinkprogress.org/1eVhC14)

THINKPROGRESS

However, it is easier to focus on just one part of this perspective

PESTLE – M consists of:



PESTLE-M as it applies to the specific focus of the enterprise that would support strategic direction

Political
Economic
Social
Technological
Legislative
Environment and
Marketing

Now add an analysis approach

- Identify the **scope** and **results** to achieve of the complete analysis not just one stage
 - Also identify the stage-by-stage results expected
- Create lists of things in the scope
- Create relationships and bounding of things in the scope
 - E.g. models like matrices, flows, lists etc.
- Apply some analytics
 - Ranking, Inference, comparison etc.
- Draw some conclusions –
 - e.g. what is the most important, most impacted least significant etc. components in each list

Prepare lists of the items of interest in each part of PESTLE-M

For example,
PESTLE Lists:

Legislative

KL	Name
	CPSC - Consumer Product Safety
	EPA - Environmental Protection Agency
	EEOC - Equal Employment Opportunity Commission
	FAA - Federal Aviation Administration
	FCC - Federal Communication Commission
	FDIC - Federal Deposit Insurance Corporation
	FED - Federal Reserve System
	FTC - Federal Trade Commission
	FDA - Food and Drug Administration
	ICC - Interstate Commerce Commission
	NLRB - National Labor Relations Board
	NRC - Nuclear Regulatory Commission
	OSHA - Occupational Safety and Health
	SEC - Securities and Exchange Commission

Social

KL	Name
	Social Commerce
	Integrated Advertising
	Geo - Targeting
	Geo - Fencing
	Social Video
	Agile Marketing
	Niche Networks
	Wearable Tech
	Vlogging
	Paid Media
	Social Networking
	Currencies of change
	Shifting Demographic Interests
	Graying society
	More Singles fewer marriages

Technology

KL	Name
	Nanotechnology
	Biotech
	Biometric
	Genetic Modification
	Autonomous Vehicles
	Quantum Computing
	3-D Printing
	Cloud Computing
	Internet of Things
	Data Analytics
	Smart Machines
	Sensor Technology
	Social media
	Industrial Robotics
	Cyber Risk

Economic

KL	Name
	Aggregate Supply
	Aggregate Demand
	Business Formations
	Consumption
	Disposable Income
	Economic Growth
	Employment
	Exports
	Firm Revenues
	Government Spending
	Gross Domestic Product
	Household Formations
	Imports
	Inflation
	Investments
	Price Adjustments

Markets

KL	Name
	Demo - Age bracket
	Demo - Income
	Geo - Americas
	Geo - EU
	Geo - MENA
	Geo - SEA
	Prod - Electronic cooking
	Prod - Gas cooking
	Prod - Refrigeration
	Prod - Home laundry
	Prod - Dishwashers
	Prod - Air Conditioning
	Prod - Heating
	Prod - Residential water Heaters
	Prod - Commercial water heaters

Next connect them to what determines your success such as

- Markets

This requires a set of touchpoint or context matrices with Markets as the row headings, e.g.

Model Name: Market Impact by Regulation [From Dimension: EXTERNAL ARCHITECTURE -- To Dimension: EXTERNAL ARCHITECTURE]

All

☐ Expand ☒ Contract Class Relations: 58 -- Density: 27.62%

To From	CPSC - Consumer Product Safety Administration	EPA - Environmental Protection Agency	FTC - Federal Trade Commission	NLRB - National Labor Relations Board	OSHA - Occupational Safety and Health	ICC - Interstate Commerce Commission	EEOC - Equal Employment Opportunity Commission	FCC - Federal Communication Commission
Demo - Age brack	●				●			
Geo - Americas (M	●	●		●		●	●	
Prod - Dishwashers	●		●	●	●			
Prod - Gas cookin	●	●	●	●	●			
Prod - Refrigeratio	●	●		●	●			
Prod - Home laund	●			●	●			
Demo - Income (M			●		●	●		
Geo - EU (Markets	●	●			●		●	
Geo - MENA (Marl		●	●					
Geo - SEA (Marke		●	●					
Prod - Electronic c	●		●	●	●			●
Prod - Air Condition		●	●	●	●			●
Prod - Heating (Ma		●	●	●	●			●

Identify PESTLE to market impacts

This requires creating a set of similar matrices and combining the frequencies into a composite view

Model Name: Economic Impact on Markets [From Dimension: EXTERNAL ARCHITECTURE -- To Dimension: EXTERNAL ARCHITECTURE]

All
Expand Contract Class Relations: 115 - Density: 40.36%

From	To	Aggregate Supply (Economic)	Business Formations (Economic)	Household Formations (Economic)	Imports (Economic)	Price Level (Economic)	Disposable Income (Economic)	Inflation (Economic)	Consumption (Economic)	Government Spending (Economic)	Exports (Economic)	Economic Gro (Economic)
Demo - Age brack												
Geo - Americas (M)												
Prod - Heating (M)												
Prod - Refrigeratio												

Model Name: Market Impact by Regulation [From Dimension: EXTERNAL ARCHITECTURE -- To Dimension: EXTERNAL ARCHITECTURE]

All
Expand Contract Class Relations: 58 - Density: 27.62%

From	To	CPSC - Consumer Product Safety (Regulation)	EPA - Environmental Protection (Regulation)	FTC - Federal Trade Commission (Regulation)	NLRB - National Labor Relations (Regulation)	OSHA - Occupational Safety & Health (Regulation)	ICC - Interstate Commerce (Regulation)	EEOC - Equal Employment Opportunity (Regulation)	FCC - Federal Communication (Regulation)
Demo - Age brack									
Geo - Americas (M)									
Prod - Electronic c									
Prod - Gas cookin									
Prod - Dishwashers									
Prod - Refrigeratio									
Prod - Home laund									
Prod - Dishwashers									
Prod - Air Condition									

Model Name: Market Impact by Social [From Dimension: EXTERNAL ARCHITECTURE -- To Dimension: EXTERNAL ARCHITECTURE]

All
Expand Contract Class Relations: 93 - Density: 41.33%

From	To	Agile Marketing (Social)	Currencies of change (Social)	Geo - Fencing (Social)	Geo - Targeting (Social)	Social Video (Social)	Social Commerce (Social)	Integrated Advertising (Social)	Social Networking (Social)	Growing society (Social)	More Singles / fewer marrieds (Social)	Niche Networks (Social)
Demo - Age brack												
Demo - Income (M)												
Geo - Americas (M)												
Geo - EU (Markets)												
Geo - MENA (Mark)												
Geo - SEA (Markets)												
Prod - Electronic c												
Prod - Air Condition												
Prod - Heating (M)												

Model Name: Technology Impacts on Markets [From Dimension: EXTERNAL ARCHITECTURE -- To Dimension: EXTERNAL ARCHITECTURE]

All
Expand Contract Class Relations: 84 - Density: 35.0%

From	To	3D Printing (Technologies)	Autonomous Vehicles (Technologies)	Biometric (Technologies)	Industrial Robotics (Technologies)	Nanotechnology (Technologies)	Sensor Technology (Technologies)	Social media (Technologies)	Mobile Computing (Technologies)	Internet of Things (Technologies)	Cyber Risk (Technologies)	Cloud Comput (Technologies)
Demo - Age brack												
Demo - Income (M)												
Geo - Americas (M)												
Prod - Dishwashers												
Prod - Heating (M)												
Prod - Residential												
Geo - EU (Markets)												
Geo - MENA (Mark)												
Geo - SEA (Markets)												
Prod - Electronic c												
Prod - Gas cookin												
Prod - Refrigeratio												
Prod - Home laund												

Row headings re the same in each matrix

Finally analyze the composite ranking of the markets most likely impacted by changes in PESTLE

Markets --> Legislative
Matrix) Economic Impact on Markets
Markets --> Economic
Matrix) Market Impact by Social
Markets --> Social
Matrix) Technology Impacts on Markets
Markets --> Technologies

INSTANCE RANK:

1 -- 1.25 -- Geo - Americas
2 -- 3.25 -- Geo - SEA
2 -- 3.25 -- Geo - MENA
2 -- 3.25 -- Demo - Income
3 -- 3.5 -- Prod - Air Conditioning
3 -- 3.5 -- Geo - EU
4 -- 3.75 -- Prod - Electronic cooking
4 -- 3.75 -- Demo - Age bracket
5 -- 4.5 -- Prod - Refrigeration
5 -- 4.5 -- Prod - Commercial water heaters
5 -- 4.5 -- Prod - Heating
5 -- 4.5 -- Prod - Gas cooking
6 -- 4.75 -- Prod - Home laundry
7 -- 5. -- Prod - Residentail water Heaters
8 -- 5.25 -- Prod - Dishwshers

Workshop Exercises:

Exercise Part 1 – Rank markets using the external environment defined by PESTLE – M

- Defines part of external architecture and links External-to-Strategic architectures

Exercise Part 2 – Identify which products are most impacted by the market rank and infer which major processes are impacted by the market rank

- Links Strategic-to-Business architectures

Exercise Part 3 – Infer which capabilities and value streams are impacted by the market rank

- Links Business architecture-to-Enterprise architecture

Exercise Part 4 – Infer which systems, processes and data are impacted by the market rank

The Blue/Red Ocean Idea

- A blue ocean is the one in which there are few competitors and you have a hot product
- A red ocean is where there are lots of products and lots of competitors and you are fighting for share
- There is a 4-Box way of looking at this

The 4 – box approach



What we will do in Exercise 1

- Start with a set of lists:
 - Social, Technology, Legislative and Economic influences, markets, products
 - Create matrices of PESTLE relationships to Markets using markets as the row headings
 - Calculate a simple composite ranking of marked based on frequency of hits
 - *First Conclusion: The top-ranked market is the one most impacted and the bottom is the least.*
 - *Analyze the 4 box for Blue versus Red ocean action*

Phrase analytic tools are used for this



Section 1 – Exercise

Framing the External Environment